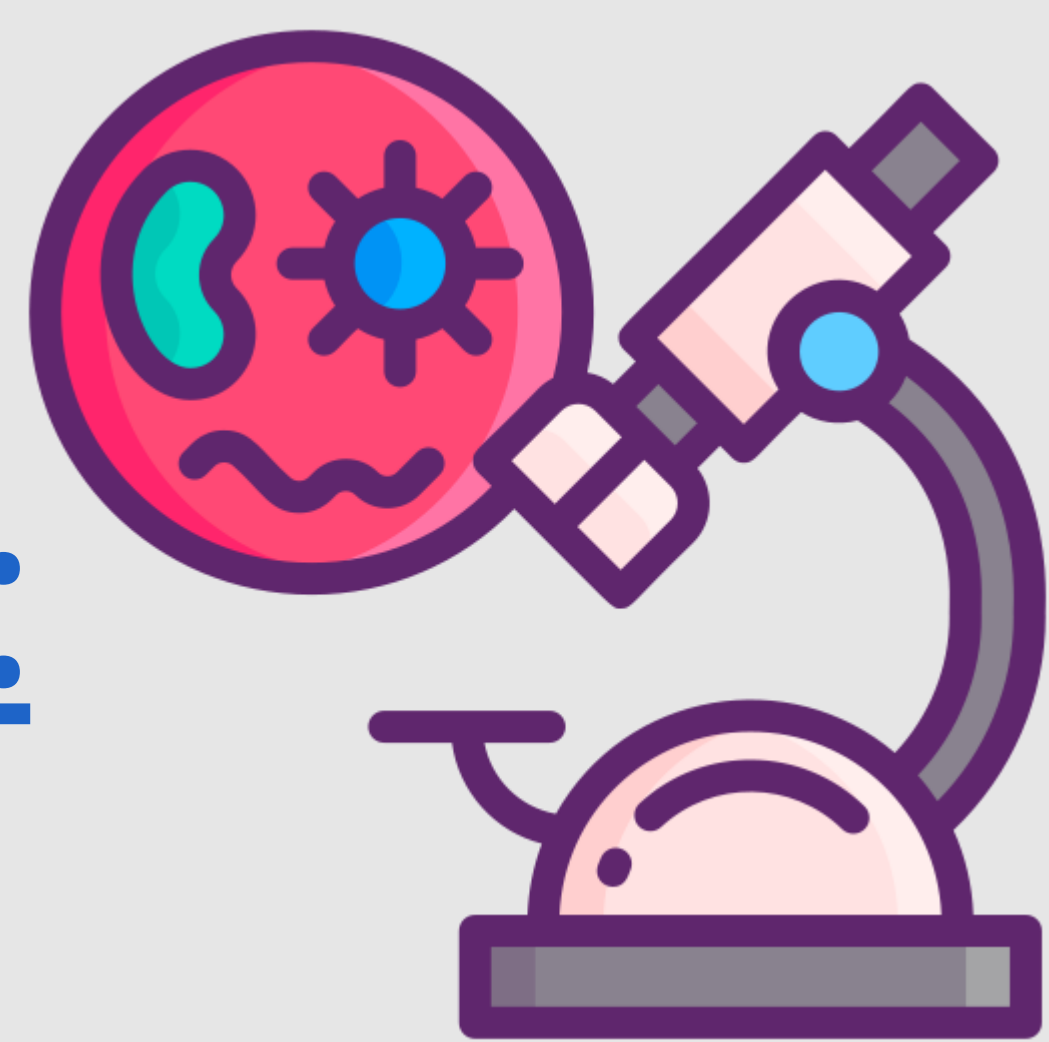




AI-POWERED QUALITY CONTROL IN BIOPROCESSING: AUTOMATION OF CRO TEST WITH COMPUTER VISION

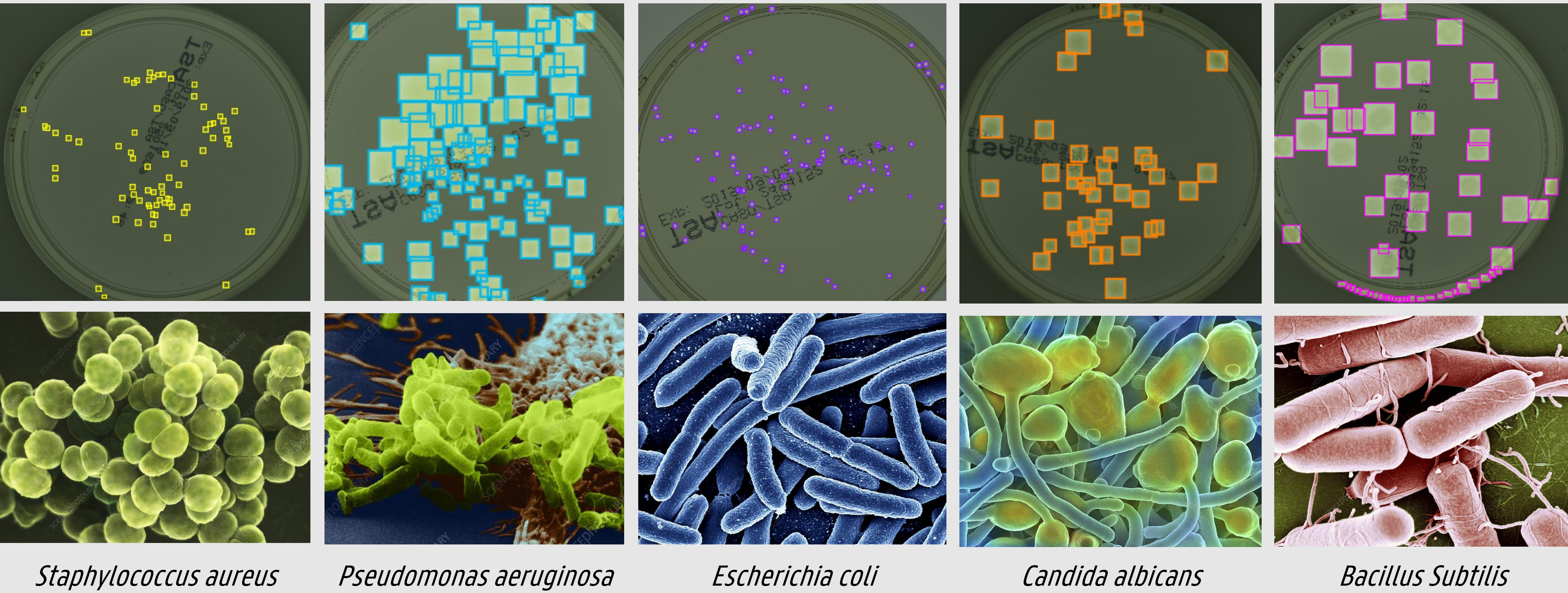
Introduction

Why the automation of bioprocessing required?

Bioprocessing of pharmaceuticals undergoes complicated steps under strict regulatory supervising to protect consumers from potential hazards. The quality control (QC) primarily focuses on compliance, consistency, and risk mitigation to quantitatively assess the microbiome and ensure the absence of harmful microorganisms in both the product and the equipment used in the process. However, the current QC system has certain limitations so that automation of QC is required based on the computer vision technique using artificial intelligence (AI).

Dataset Selection

Images of microbial colonies on petri-dish in laboratory that were annoated by sequencing is used for the model fine-tune. The following set of pictures shows annotation and the actual observation of each microbes under SEM. Image source: Roboflow dataset, Science photo library (SEM photo)



Finetune yolov8 model

Total 991 images (jpg) and labels (txt) used for fine-tuning the yolov8 model, with guide of data.yaml file.

Train dataset
695 images, 695 labels

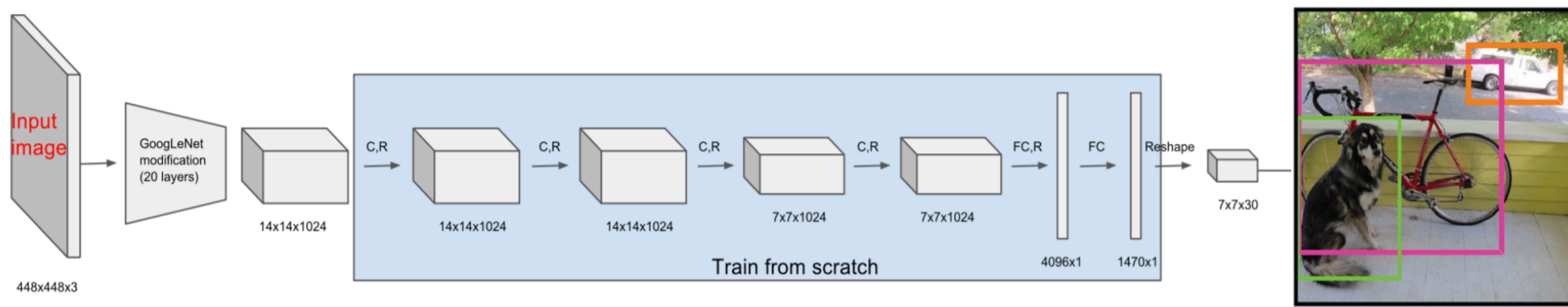
Validation dataset
198 images, 198 labels

Test dataset
98 images, 98 labels

*A pre-annotated dataset was taken for this study.

Model Description

For object-detection to automize QC in bioprocessing, the one step object detection model yolo is used. In long-term view, since a real-time detection should be the final goal, the fastest model yolov8n is selected for fine-tuning.



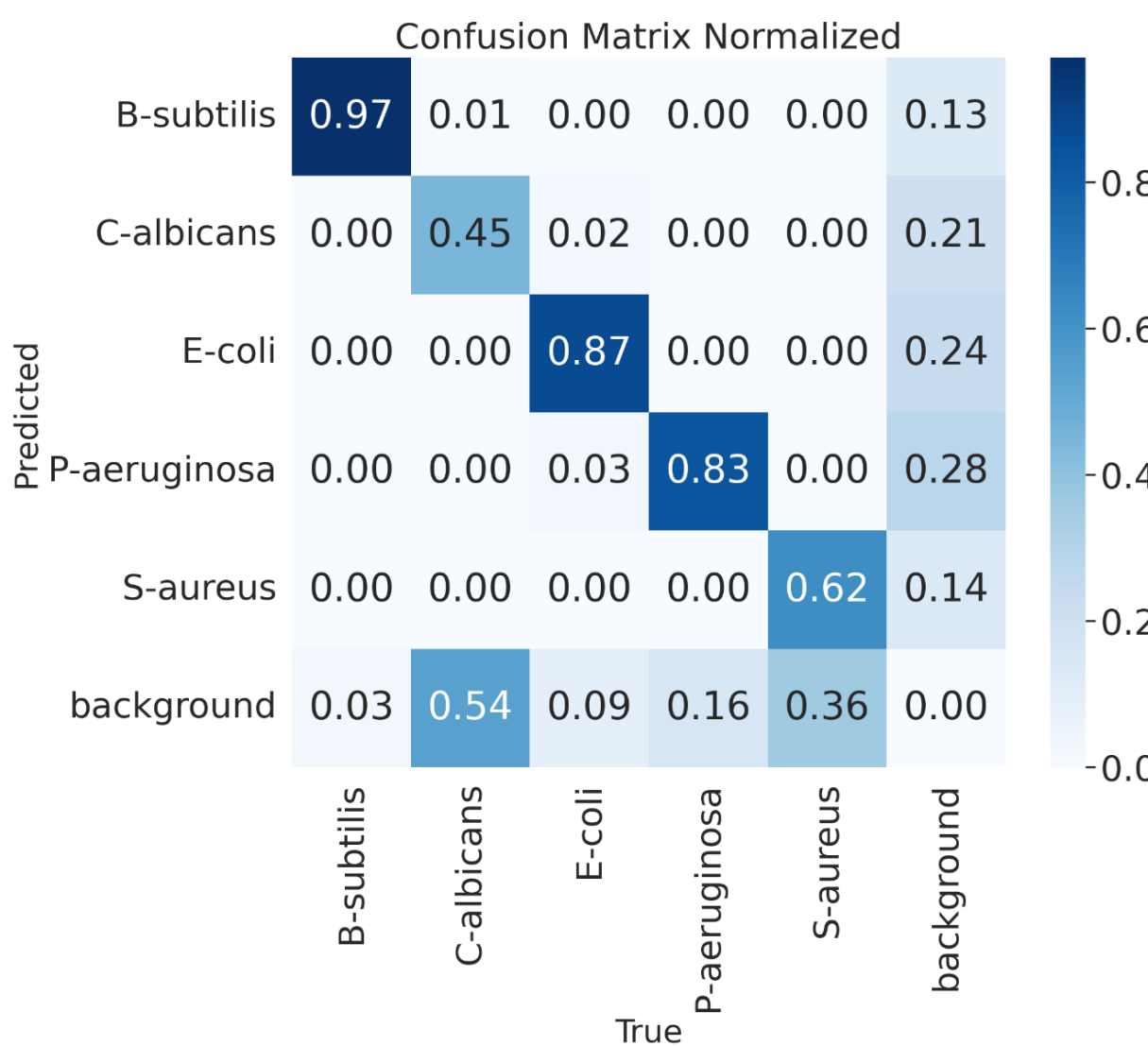
Result

Test metric values of the best model, by the variance of the number of epoch.

Epoch #	Precision	Recall	F1 score
50	0.770	0.566	0.652
100	0.879	0.611	0.721
150	0.957	0.593	0.732
200	0.955	0.602	0.738

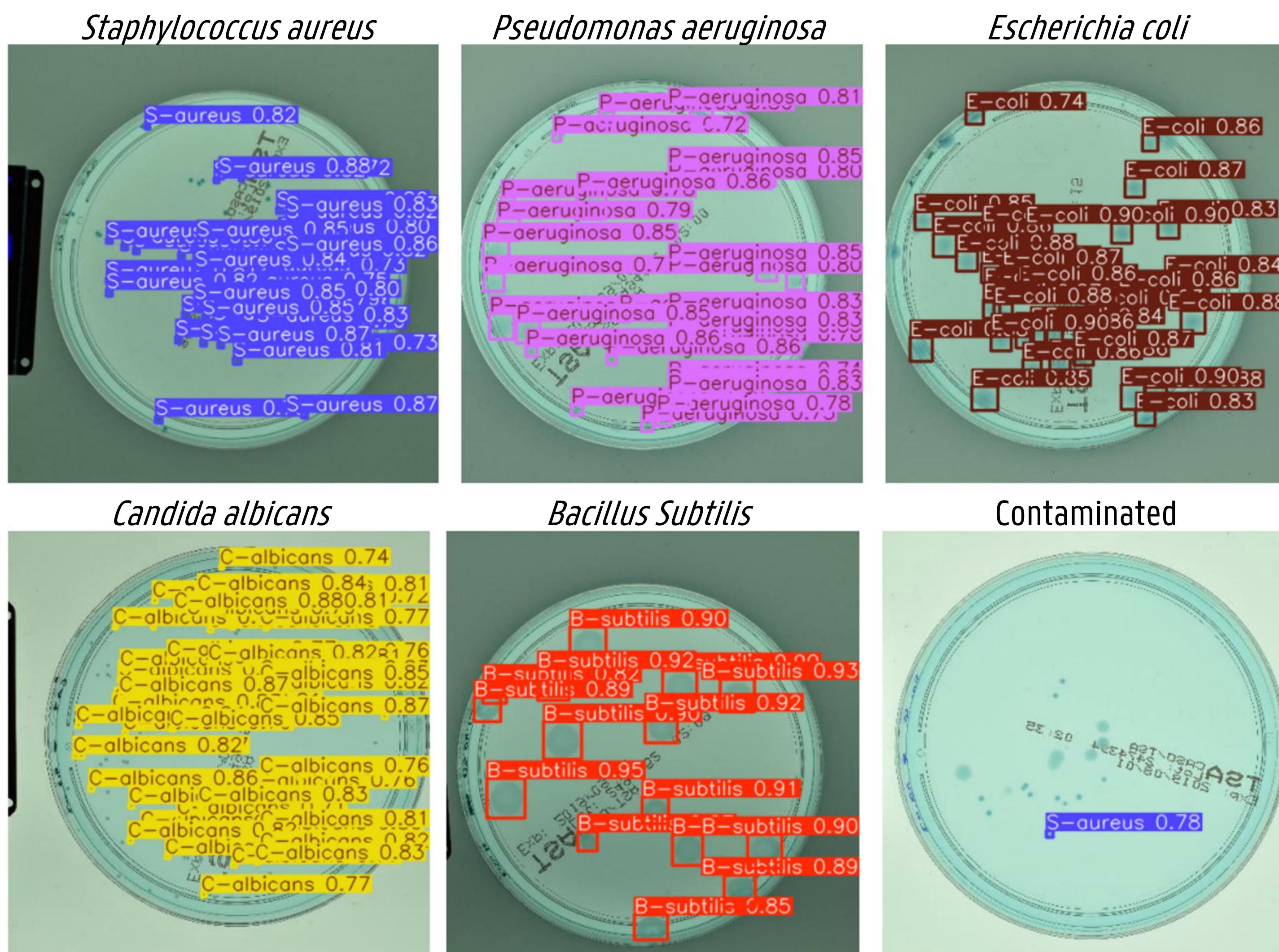
Conclusion

- The detection performance is reasonable after 200 epoch, according to metric values and the inferences
- Colonies that has relatively small size (*S. aureus* and *C. albicans*) are often omitted
- Additional experiments required to find the best threshold while balancing precision & recall value and that could improve F1 score significantly



Test Inferences

Model inferences on unseen test dataset. For more, please visit the Kaggle notebook result page.



Further Study Ideas

- Real-time detective model training for rt-QC in bioprocessing using video dataset
- Microorganism detection in oral/colon/stomach epithelial cells with larger dataset
- Model training for laboratory usage (creation of pretrained model for biolab)
- Augmentation technique that fits to the model of micro-organism detection
- Aevelopment of the lightweight model with real time detective finetuning

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